

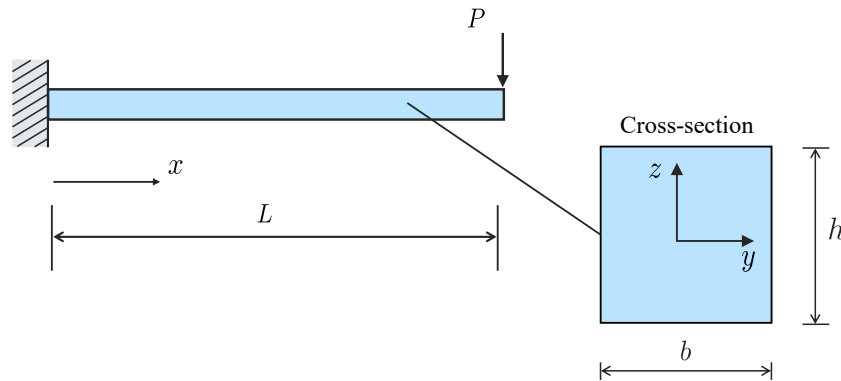
Computer assignments 1 & 2
Mechanics of solids
-solid and structural mechanics

by

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Computer assignment 1 A clamped beam shall be used to carry a person. Assume that the beam has a rectangular cross section with height h and width b . Further, the beam is made of steel and is assumed to behave linear elastic with Young's modulus E and Poisson's ratio ν . For the worst case that the person is standing at the right end of the beam (corresponding to force P in the figure below) perform the following investigations.



For two lengths of the beam, $L_1 = 2$ m and $L_2 = 0.25$ m, do the following:

1. Determine the deflection $w(x)$ and, in particular, maximum end deflection $w(x = L)$ of the beam. Account for the self-weight of the beam.
2. Compute the normal stress at the level $z = -h/2$, i.e. $\sigma_{xx}(x, -h/2)$. Also, compute the von Mises effective stress at the same level i.e. $\sigma_{eM}(x, -h/2)$. Will the beam yield according to this criterion?

Solve the problems using

- Euler-Bernoulli's beam theory.
- Timoshenko's beam theory.
- Abaqus (Solve the general equilibrium equations assuming plane stress). Apply the load as a distributed load P/h over the right side of the beam.
- Calfem toolbox in Matlab/Python (use meshing routine that is available on Canvas). Also here assume plane stress and apply the load as a distributed load. Solve CA2 before this part of CA1.

Compare and discuss the results obtained for the different models. To be specific:

- Perform a mesh convergence study (check that $w(x)$ converges "sufficiently") for Abaqus and Calfem.
- Plot in a graph $w(x)$ and in another graph $\sigma_{xx}(x, -h/2)$ for the different solution methods/models. Do this for both beam lengths. Discuss how well Euler-

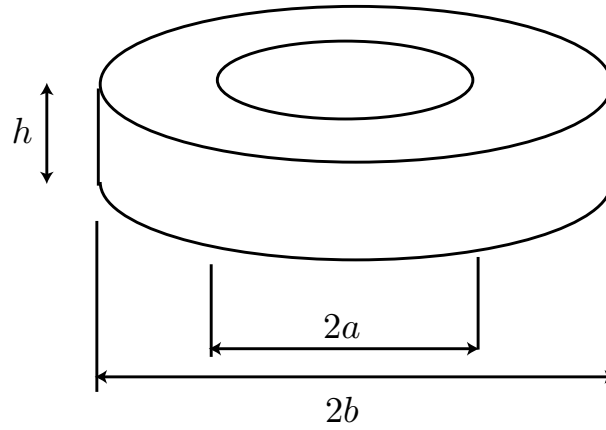
Bernoulli and Timoshenko models fit to the FEM solutions for the different lengths.

- For Abaqus, plot how the horizontal displacement $u_x(x = L/2, z)$ varies with z . Is the assumption of the beam kinematics that it varies linearly with z reasonable?
- For Abaqus, can you identify any stress concentrations? How can you change the design of the beam to avoid this/these stress concentration/-s?

Material and load data: $E = 200$ GPa, $\nu = 0.3$, $\sigma_y = 450$ MPa, $\rho = 7800$ kg/m³, $g = 9.81$ m/s², $m = 120$ kg ($P = m g$)

Cross-sectional data: $h = b = 0.05$ m, shear correction factor $K = 5/6$ (for the Timoshenko model).

Computer assignment 2 For a thin axisymmetric disc (geometry and loading) compute stresses and displacements using analytical derivations and FEM.



Assume the following boundary conditions: pressure on the outer boundary $\sigma_{rr}(r = b) = -p$ and clamped at the inner boundary $u_r(r = a) = 0$. Determine the displacement $u_r(r)$ and the normal stress σ_{rr}

- By solving the differential equation for u_r analytically. Assume a constant thickness $h = h_0$ and plane stress condition.
- By applying FEM and writing your own code in Matlab/Python (it is recommended to use Calfem subroutines and variables). Compare with the solution obtained in (a) with constant thickness $h = h_0$.
- Use the same code as in (b) but assume a varying thickness $h = h_0 (r - a)/(b - a) + h_0$. Compute $u_r(r)$.

Given data: $a = 100$ [mm], $b = 200$ [mm], $h_0 = 20$ [mm], $p = 100$ [MPa], $E = 210$ [GPa], $\nu = 0.3$